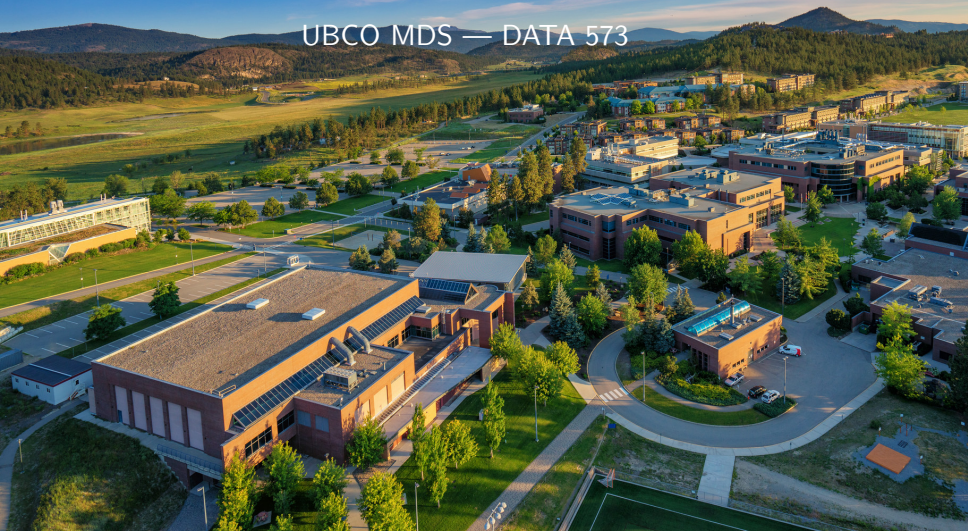


Distance Measures

UBCO MDS — DATA 573



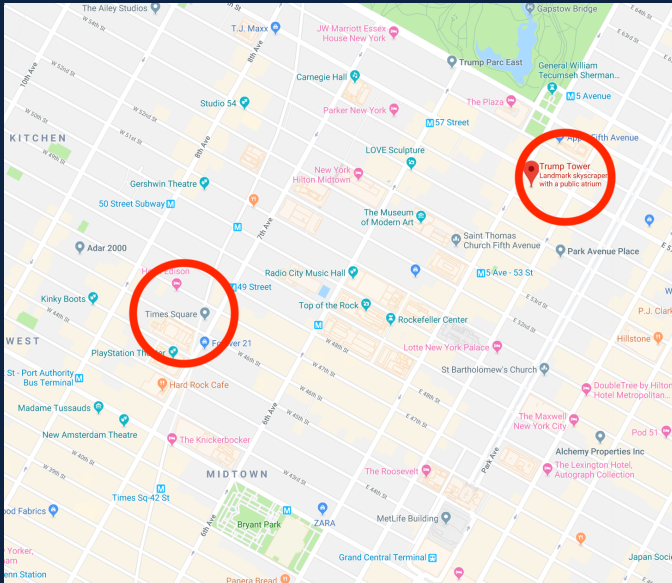


- ▶ Prior to diving deep on unsupervised methods, it's important to note that the vast majority are based on distance calculations (of some form)
- ▶ Distance **seems** like a straightforward idea, but it is actually quite complex....

Distance Motivation



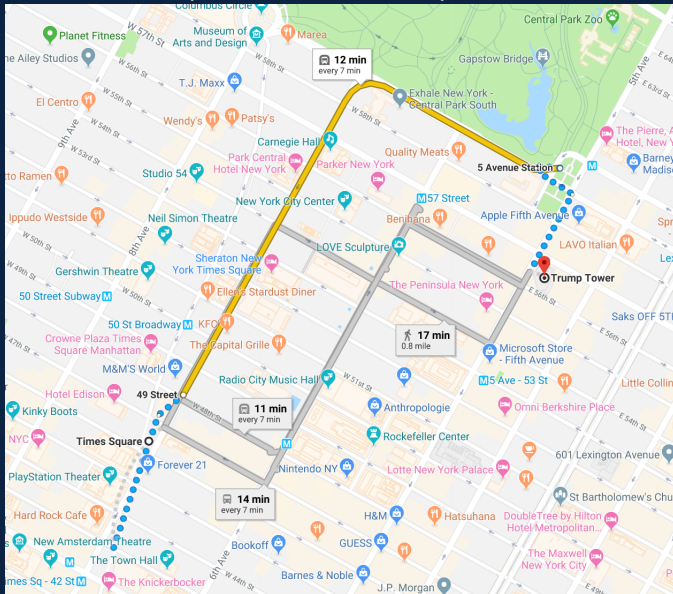
► Distance from Trump Tower to Times Square?



Distance Motivation



► Distance from Trump Tower to Times Square?



Distance Motivation



► Distance from Trump Tower to us?



- ▶ To begin, we'll consider that all predictors are numeric.
- ▶ First, the main distance that you know of...
- ▶ Euclidean distance between observations i and j simply as

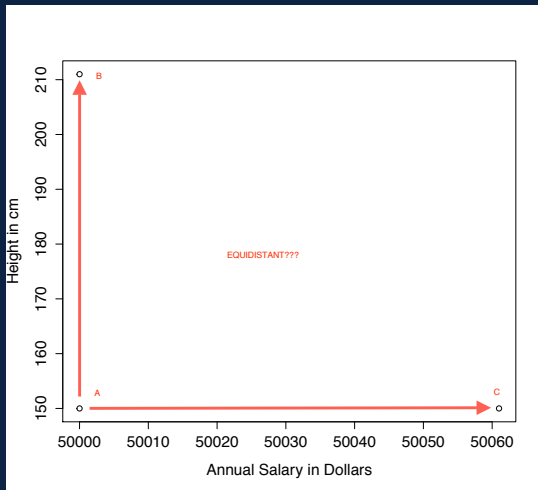
$$d_{ij}^{\text{EUC}} = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

is inappropriate in many settings.



- ▶ And now, why the main distance that you know of is immediately problematic...
- ▶ Consider the following measurements on people:
 - ▶ Height (in cm)
 - ▶ Annual salary (in \$)
- ▶ A \$61 difference in annual salary would be considered a miniscule difference, whereas a 61 cm difference in height (approx 2 feet) would be substantial!

Defining Distances





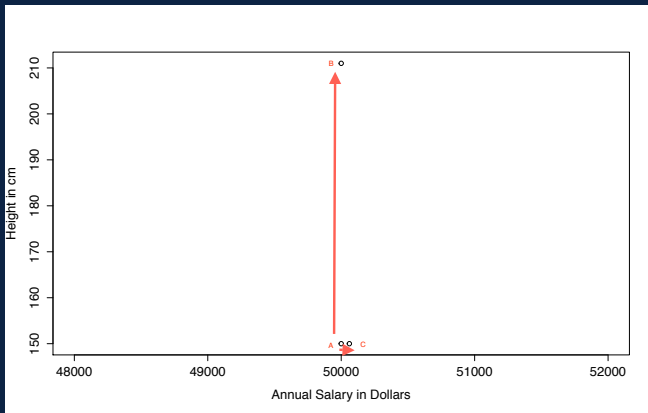
- Scale the data to have mean 0, variance 1 via

$$z_{ik} = \frac{x_{ik} - \mu_k}{\sigma_k}$$

- Then we can define pairwise distances as

$$d_{ij}^{\text{STA}} = \sqrt{\sum_{k=1}^p (z_{ik} - z_{jk})^2}$$

Defining Distances



Manhattan (or City-Block) Distance



- ▶ **Manhattan** or **city-block** distance merely measures distance as though one could only travel along the axes.
- ▶ So pairwise distance are calculated as

$$d_{ij}^{\text{MAN}} = \sum_{k=1}^p |x_{ik} - x_{jk}|$$

- ▶ Similarly, one might want to consider the standardized form

$$d_{ij}^{\text{MANs}} = \sum_{k=1}^p |z_{ik} - z_{jk}|$$



- ▶ **Mahalanobis distance** takes into account the covariance structure of the data.
- ▶ Recall, we saw this distance crop-up in our discussion of LDA/QDA.
- ▶ It is easiest defined in matrix form

$$d_{ij}^{\text{MAH}} = (\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{\Sigma}^{-1} (\mathbf{x}_i - \mathbf{x}_j)$$

- ▶ Note that Mahalanobis distance is a generalization of standardized euclidean distance (you can think of it as standardizing by both variance and covariances).

When to Standardize?



- ▶ This brings us back to a question students struggle with...when should we use a standardized measure, and when should we not?
- ▶ General rule of thumb: it is **often** a good idea, and usually necessary if measurements are on vastly different units.
- ▶ Another way to put it: unless you have a **very good reason** to believe that higher variance measures should be weighted heavier in your unsupervised method, then you should standardize in some form.
- ▶ FYI: some unsupervised methods are scale invariant, in which case you don't have to worry about it.

- ▶ What if some/all the predictors are not numeric?
- ▶ We'll start with the binary case through the following example

President	Democrat	Governor	VP	2nd Term	From Iowa
GW Bush	0	1	0	1	0
Obama	1	0	0	1	0
Trump	0	0	0	1	0
Biden	1	0	1	0	0

Matching Binary Distance



President	Democrat	Governor	VP	2nd Term	From Iowa
GW Bush	0	1	0	1	0
Obama	1	0	0	1	0
Trump	0	0	0	1	0
Biden	1	0	1	0	0

- Equivalent to Manhattan/City-block,

$$d_{ij}^{\text{MAT}} = \# \text{ of variables with opposite value}$$

- Example GW/Bush vs Obama:

$$d_{12}^{\text{MAT}} = |0 - 1| + |1 - 0| + |0 - 0| + |1 - 1| + |0 - 0| = 2$$

- Often divided by p for simple interpretation

$$d_{12}^{\text{MAT}_p} = \frac{2}{5} = 0.4$$

Asymmetric Binary Distance



President	Democrat	Governor	VP	2nd Term	From Iowa
GW Bush	0	1	0	1	0
Obama	1	0	0	1	0
Trump	0	0	0	0	0
Biden	1	0	1	?	0

- ▶ But are the 0's in "From Iowa" really matches?
- ▶ Asymmetric binary:

$$d_{ij}^{\text{ASY}} = \frac{\# \text{ of } 0\text{-}1 \text{ or } 1\text{-}0 \text{ pairs}}{\# \text{ of } 0\text{-}1, 1\text{-}0, \text{ or } 1\text{-}1 \text{ pairs}}$$

- ▶ Example GWBush vs Obama:

$$d_{12}^{\text{ASY}} = \frac{2}{3} = 0.667$$

- ▶ For nominal variables, a common measure is essentially standardized matching once again
- ▶ If $u = \#$ of variables that match between observations i and j and $p = \#$ of variables then

$$d_{ij}^{\text{NOM}} = 1 - \frac{u}{p}$$

- ▶ Often data has a mix of variable types. Gower's distance is a common choice for computing pairwise distances in this case.
- ▶ Basic idea is to ensure that each variable's contribution to total distance is standardized between 0 and 1, and then sum them up
- ▶ Gower's distance:

$$d_{ij}^{\text{GOW}} = \frac{\sum_{k=1}^p \delta_{ijk} d_{ijk}}{\sum_{k=1}^p \delta_{ijk}}$$

where

- ▶ $\delta_{ijk} = 1$ if both x_{ik} and x_{jk} are non-missing (and note that a 0-0 binary match is often counted as “missing” to match asy-binary)
- ▶ $\delta_{ijk} = 0$ otherwise.
- ▶ and d_{ijk} depends on variable type...

Gower's Distance



- ▶ Quantitative numeric

$$d_{ijk} = \frac{|x_{ik} - x_{jk}|}{\text{range of variable } k}$$

- ▶ Qualitative (nominal, categorical)

$d_{ijk} = 0$ if obs i and j agree on variable k , 1 otherwise

- ▶ Binary (asymmetric)

$d_{ijk} = 0$ if obs i and j are a 1-1 match, 1 otherwise

- ▶ Example on board.



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